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Process Quality Requirements (LAH)

Tire-Wheels Assembly Project: VW 326

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B Introduction

B.1 Objective

The objective of this document is to assure product quality through a robust and reliable process during the selection procedure of the best suppliers to guarantee quality and quantity of the product during the life span of the project. This document is considered to apply the lessons learned from previous projects. In addition, it gives the supplier and the costumer the confidence that all requirements and prerequisites of the customer are taken into consideration.

B.2 Content of the LAH and areas for validation

It is the responsibility of the supplier that the completed content of this document and evidence of compliance with the requirements listed below will be considered for future nomination and must be presented on the QTR evaluation.

The Q-LAH could be pushed by all areas, since this is a process that involved several sectors.

This document does not substitute any of the manuals: Formel Q, Formel-Q Konkret, VDA 6.3. TE-LAH

C General

C.1 Background

The aim of the Q-LAH is that the offer, based on the following requirements, is checked feasibility on the four items of the Q-LAH:

- 1) Development and manufacturing capabilities
- 2) Raw Material
- 3) Process development
- 4) Finish Goods

The quoting supplier is must make all necessary documents available before the nomination.

C.2 Procedure

The Q-LAH available for Prio 1 parts will be send by Quality Acquisitions to the purchase department. The supplier shall be notified by the purchase department through RFQ and this Q-LAH documentation must be considered for submit a quotation.

All the points present in this document and the evidence are necessary to have feasible results on a QTR evaluation.

Volkswagen of Mexico holds the right to request the data for future reference, if necessary.

C.3 Results

If during the evaluation of the QTR, based on the information submitted is displayed that the provider does not meet the requirements of this Q-LAH, the grade will be Not Feasible and this result influence the procurement process and the final nomination.

1 Development and manufacturing capabilities

The supplier has to fulfil in line with the technical requirements in the area of process development. The below listed requirements shall cover the main topics of the capabilities for the project development.

- Design and Manufacturing knowledge of all Processes involved on Tire-Wheel assembly.
- The Volkswagen Design Guidelines and Standards must be observed Technical Development Specialist in charge leads this issue.
- The valid documents are available on the B2B supplier platform of the Volkswagen Group at <http://www.vwgroupsupply.com>.
- Tooling and Manufacturing equipment responsibility must be followed as per RFQ.
- The relevant legal requirements of the target markets must be fulfilled. All testing and activities management to get certifications and registration numbers when needed must be performed by the supplier.
- Supplier must inform and confirm the Manufacturing Facility DUNS number beside the Commercial address DUNS number.

2 Raw material

Assembly components must be controlled with a receiving inspection and in line check points procedures; supplier must follow a non-conforming material procedure to give proper disposition timely to all rejected parts/components in agreement with Q-VWM specialist in charge. All defect acceptance criteria and inspection methods must follow the corresponding VW specification (eg. Aluminium Wheels must meet PV6009).

An example of the non-conforming material management procedure is available upon request.

The direct communication between assembler and component manufacturer is a plus to assure Q issues management. This communication must be performed in agreement with Q-VWM specialist in charge.

3 Process development (each step)

It is the responsibility of the supplier to create a conclusive manufacturing concept, which guarantees and controls all part specific characteristics to be correct. If already available equipment and strategies are used an additional point of lessons learned has to be included and presented.

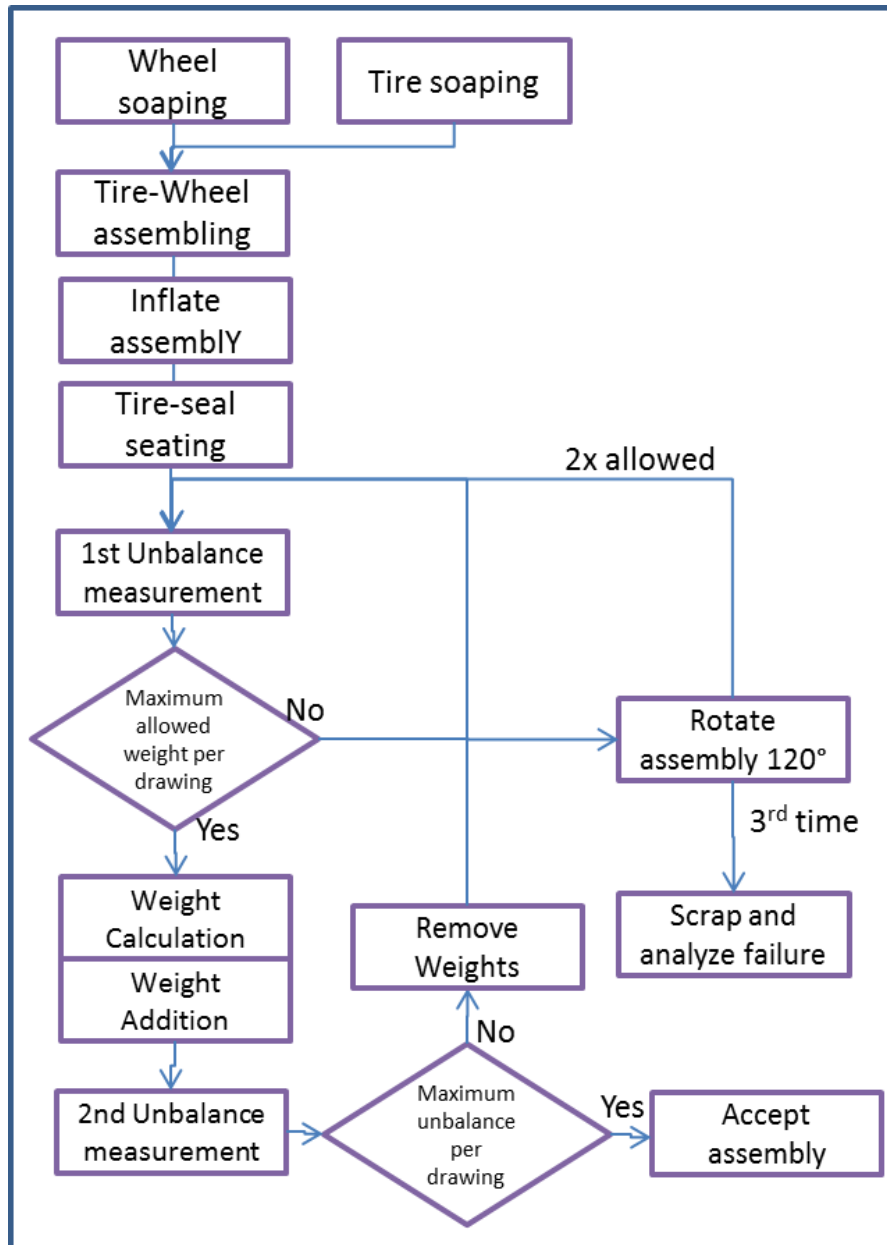
The requested documents are: a) LAYOUT, b) MANUFACTURING PROCESS DESCRIPTION and c) CONTROL CHECK POINTS AND EQUIPMENT.

3.1 Information requested by Processes:

- **Raw Material Control.**
 - First Input First Output material disposition practice.
 - Tight lots control for material status, only released material sent to production.
- **Minimum Required Process Sequence. (TE-LAH must be accomplished too)**
 - Valve addition to wheel.
 - Feed conveyor with Wheel and Tire – first inspection.

- Soap Wheel – Soap Tire.
- Automatic assembly.
- Inflate assembly – Pressure control (44 +/- 2 psi)
- Tire seal seating (Batanado)
- Assembly balance measurement and weight calculation.
- Weight addition.
- Assembly balance certification 8 grs-cm. Max. by plane.
- Final inspection and Q-Audit.

At least the processes on the following process flow chart are required:



The preferred 2nd unbalance measurement is 100% of production, when no equipment installed or agreed then an audit process can be allowed at least 5 part every hour are required and a Cpk of 1.33 is requested.

4 Finished goods

The supplier is responsible, to show a checking concept for all critical / functional product characteristics. The capability for the control resources has to be proven.

4.1 Control Methods

- ID Marks- As per drawing.
- Function testing for series production control (Test frequency is to be agreed with the responsible BTV).
- Total imbalance Maximum 8 gcm. Per drawing.
- Air pressure. Per drawing.

- Final inspection area.
 - Follow VWM inspection standard methods.

4.2 Material handling and transportation.

- All internal handling racks including quarantine area must be documented and must be sufficient to avoid damages to the product.
- The final shipping packing method and materials must be agreed with VW Logistics department.
- Supplier is the final responsible of the integrity of the part when shipping and transport to the customer at Use Point and must have ready an alternate packing method to avoid supply failures, all material and packing methods must assure the integrity of the product and the safety of the personnel handling it.

5 Services

The following services shall be provided in order to achieve the quality objectives during the development of the project, SOP and continued standard production:

- A complete Organizational chart should be presented for further QTR evaluations. The supplier must consider exclusive team for this project (project leader, quality and engineering persons), guaranteeing that the same team will follow up the development until Note 1 of product and final approval of manufacturing site are achieved.
- The Tier 1 is responsible to control and register the current component status during the pre-production phase in consultation with the responsible VWM Quality Assurance Department.

The supplier is responsible of register every technical or process change that impacts the product, with a new generation level in agreement with the VW Quality responsible (BTV). This has to be registered in the history part document and physically on the parts.

Ref. Formel Q New Parts Integral, Quality Documentation for pre-series phase.

- Quality samples for release & parts for special events must be provided by supplier, including the evaluated parts & set components needed for the evaluation in agreement with the Q specialist. The parts must be free of charge to Volkswagen.
- Volkswagen reserves the right to sampling scopes. A reduction in the sampling scope can be agreed for components already sampled by VW, in agreement with the VW Quality Assurance for Purchased Parts responsible (BTV).
- Samples requested by VW quality assurance must come with dimensional report (exceptions must be agreed with BTV). In case of a Photogrammetry report (Scanning) needed, the supplier must provide it.

- For the entire pre-series events the supplier must assure the latest engineering level for the final product and components. Amount and level must be in agreement with VW logistic, Quality, Engineering and purchasing departments.
- For preseries and series the supplier must consider a qualified resident Engineer in Volkswagen facilities, for solving and holding problems. From standard production onwards, the supplier must have a 24 hour available QA officer, skilled in the supplier process for attendance at the vehicle audit, cubing, production assembly, etc.
- If quality problems occur, the supplier shall initiate immediately containment actions. Written comments on damage analysis reports (e.g. in the form of 8-D Reports) shall be provided within 24 hours.
- Adequate analysis options shall be provided. The supplier must deliver on time and amount requested by VW. If needed the supplier must acquire additional sources to fulfil VW requirements.
- VWM reserves the right to debit the suppliers in case of planned activities have not been accomplished on time scheduled. (ex. 2 day production, facility visits, follow-ups of claims and other services).